

Noise Type Classification Using Time- and Frequency-Domain Audio Features and Machine Learning

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1 Student Information

Table 1: Student Information

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GitHub Project Link	https://github.com/lzhua020623/elec5305-project-550531858

2 Project Overview

This project aims to develop a Python-based system that classifies different types of environmental noise using time-domain and frequency-domain audio features together with machine learning. Environmental sounds such as rain, wind and vehicle engines have different patterns in their waveforms and frequency spectra. Accurate noise classification can support applications such as environmental monitoring, smart devices and safety systems by helping these systems automatically understand surrounding acoustic conditions. The project will investigate whether these differences can be represented by measurable audio features and used to identify the noise type automatically.

The main goal is not only to achieve good classification accuracy, but also to study which audio features are most useful for distinguishing different noise types. Features such as zero-crossing rate, energy, spectral centroid, spectral roll-off and Mel-frequency cepstral coefficients (MFCCs) will be extracted from audio recordings. These features will then be used as inputs to a machine learning classifier. By comparing different groups of features, the project will examine how time-domain and frequency-domain information affect classification performance.

3 Background and Motivation

Environmental sound classification is widely used in urban monitoring, security systems, and smart devices. Existing research indicates that both manually designed audio features and deep learning can be applied to this task. Salamon, Jacoby, and Bello proposed the UrbanSound dataset and its classification framework[4], while Piczak proposed the ESC-50 dataset and used features such as MFCC and zero-crossing rate as classification baselines[2]. Subsequent research has primarily focused on improving classification performance through convolutional neural networks and data augmentation[1][3].

This project does not focus on more complex models but rather explores how audio features from different categories contribute to noise classification. Time-domain features, frequency-domain features, and MFCCs will be compared using the same machine learning classifier to more intuitively reveal which signal features help distinguish between different noise types.

4 Proposed Methodology

This project will be implemented in Python, primarily using NumPy, SciPy, librosa, scikit-learn and Matplotlib. The ESC-50 dataset will be used, and five types of environmental noise with distinct characteristics will be selected, such as rain, wind, sirens, engine noise and vacuum cleaner noise. Limiting the experiment to five classes keeps the project manageable while still providing enough variation to compare the behavior of different audio features.

The audio samples will be converted into a consistent format before feature extraction. Basic preprocessing, such as amplitude normalization, will be applied where needed. Each audio recording will also be divided into short frames so that features can represent local changes in the signal rather than only describing the whole recording with a single value.

Time-domain and frequency-domain features will then be extracted. Time-domain features will include signal energy and zero-crossing rate, which describe properties directly from the waveform. Frequency-domain analysis will be used to obtain features such as spectral centroid and spectral roll-off, which describe how signal energy is distributed across different frequencies. MFCCs will also be extracted because they provide a compact representation of the spectral shape of an audio signal.

The experiment will compare four feature sets: time-domain features, frequency-domain features, MFCCs, and a combination of all selected features. The extracted features will be organized into numerical feature vectors and standardized before classification so that features with different numerical ranges do not dominate the model simply because of their scale.

A Support Vector Machine (SVM) will be used as the main classifier throughout the experiment. Keeping the classifier fixed allows the comparison to focus on the effect of the audio features rather than differences between multiple machine learning algorithms. The dataset will be separated into training and eval-

uation data while maintaining a balanced representation of the selected noise classes. The SVM will be trained on the extracted features from the training data and then used to predict the noise type of unseen audio samples.

Classification performance will be evaluated using accuracy, precision, recall and F1-score. A confusion matrix will also be generated to show which noise types are correctly identified and which are commonly confused. The same evaluation process will be applied to each feature set so that their results can be compared under consistent conditions.

5 Expected Outcomes

The project aims to produce a functional Python noise classification prototype capable of reading environmental audio, extracting audio features, and predicting the corresponding noise type.

The experiments are expected to show how classification performance changes when different feature sets are used. Rather than assuming that one type of feature will perform best, the results will be used to examine whether time-domain features, frequency-domain features, MFCCs or their combination provide the most useful representation for the selected noise types.

The final deliverables will include Python source code, experimental results, evaluation metrics, visualizations, and GitHub project documentation.

References

- [1] Karol J. Piczak. Environmental sound classification with convolutional neural networks. In *2015 IEEE 25th International Workshop on Machine Learning for Signal Processing (MLSP)*, pages 1–6, 2015.
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- [3] Justin Salamon and Juan Pablo Bello. Deep convolutional neural networks and data augmentation for environmental sound classification. *IEEE Signal Processing Letters*, 24(3):279–283, 2017.
- [4] Justin Salamon, Christopher Jacoby, and Juan Pablo Bello. A dataset and taxonomy for urban sound research. In *Proceedings of the 22nd ACM International Conference on Multimedia*, pages 1041–1044, 2014.